

Question number	Answer	Mark
8(a)(i)	The only correct answer is D blood type O <i>A is incorrect because A antigens are not present on red blood cells of humans with blood group B or O</i> <i>B is incorrect because B antigens are not present on red blood cells of humans with blood group A or O</i> <i>C is incorrect because A antigens are not present on red blood cells of humans with blood group B or O and B antigens are not present on red blood cells of humans with blood group A or O</i>	(1)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	An explanation that includes four of the following points: • B antigens are recognised as foreign antigens (1) • and therefore initiate an (humoral) immune response (1) • credit details of humoral immune response (1) • resulting in antibodies released by plasma cells (1) • credit consequence of humoral immune response (1)	e.g. opsonisation / agglutination / destruction of RBCs (in liver / spleen / by phagocytes / formation of memory cells	(4)

Question number	Answer	Additional guidance	Mark
8(b)(i)	An explanation that includes two of the following points: - they {reduce / destroy / prevent the growth of / prevent the infection of} pathogenic bacteria (1) - because they compete for {nutrients / named nutrient / space} (1) - produce {toxins / chemicals} (that destroy pathogenic bacteria) (1)	ACCEPT {foreign / other} bacteria / pathogens IGNORE food ACCEPT produce vitamin K	(2)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	An answer that includes the following points: (because the bacteria can) {absorb / use} sugar for respiration (1) - to produce ATP (for the bacteria) (1)	ACCEPT glucose for sugar throughout ACCEPT {it / they} to mean bacteria	(2)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	An explanation that includes two of the following points: • there will be no (foreign) antigens on the red blood cells (1) • the immune response will not be triggered (1) • therefore this blood can be used in any transfusion (if no antigens present)	ACCEPT antigens removed from the red blood cells red blood cells will not be recognised as {foreign / non-self} ACCEPT can be used in a transfusion as will not be rejected blood will act like {group O blood / universal donor}	